

Code Review Log - CTI E-Commerce Project

Last Updated: 2026-05-03
Project: CTI E-Commerce Platform (Frontend + Backend)
Reviewers: Development Team

Review Summary

Date	Branch	Feature	Reviewer	Status	Notes
2026-05-03	main	Mobile WebViewApp	Team	Merged	Jetpack Compose, Navigation integration
2026-05-01	main	Desktop Auth	Team	Merged	Login flow, role-based access control
2026-04-29	main	Cart Management	mrodriguezul	Complete	Context API, localStorage sync
2026-04-25	dev	State Management Setup	Team	Approved	Redux/Context planning
2026-04-20	dev	Type Definitions	Team	Approved	Core interfaces, Product/Cart types

Recent Code Review Details

Review #5: Mobile WebViewApp Integration

Branch: mobile/fet/main-web-app
Date Reviewed: 2026-05-03

Commit: 3340dd0

Changes Reviewed:

- ☒ Jetpack Compose setup for Android WebView
- ☒ Navigation component integration
- ☒ WebView bridge implementation
- ☒ Lifecycle management

Observations:

- Clean separation of concerns
- Proper resource cleanup in lifecycle
- WebView configuration follows Android best practices
- No security issues detected

Approved By: Development Team**

Review #4: Desktop Authentication System

Branch: desktop/fet/Authentication

Date Reviewed: 2026-05-01

Commit: a6a419b

Changes Reviewed:

- ☒ LoginController unit tests (100% pass)
- ☒ Authentication flow (JWT-based)
- ☒ Role-based access control (RBAC)
- ☒ Secure password handling with bcrypt
- ☒ User stories updated

Code Quality:

- **Coverage:** 85% unit tests
- **Linting:** 0 errors
- **Type Safety:** TypeScript strict mode
- **Security:** OWASP top 10 compliance checked

Approved By: Development Team**

Review #3: Cart State Management Implementation

Branch: e-comm/feat/cart-management

Date Reviewed: 2026-04-29

Commit: Tip of e-comm/feat/cart-management

Changes Reviewed:

-  CartContext implementation (5 Acceptance Criteria met)
-  localStorage integration with sync
-  addToCart() - with stock validation
-  updateQuantity() - with boundary checks
-  removeFromCart() - clean removal
-  clearCart() - full reset
-  totalItems & totalAmount computed properties

Implementation Quality:

- **Pattern:** React Context API (optimal for this scope)
- **Error Handling:** Validation on stock, error thrown on invalid qty
- **Storage:** Persistent via localStorage
- **Performance:** Memoization of context value
- **Testing:** Integration tests with mock data

Code Standards Met:

-  Props validation
-  Type safety (TypeScript)
-  Error messages user-friendly
-  No console errors
-  Follows project conventions

Approved By: mrodriguezul**

Frontend Code Quality Metrics

Metric	Target	Current	Status
Test Coverage	70%	68%	⚠️ Monitor
ESLint Errors	0	0	✅ Pass
Type Check Errors	0	0	✅ Pass
Build Warnings	<5	2	✅ Pass
Accessibility (WCAG 2.1)	AA	AA	✅ Pass

Backend Code Quality Metrics

Metric	Target	Current	Status
Test Coverage	75%	72%	⚠️ Monitor
ESLint Errors	0	0	✅ Pass
Type Check Errors	0	0	✅ Pass
Security Audit	Pass	Pass	✅ Pass
API Documentation	100%	95%	⚠️ In Progress

Common Review Findings

✅

Best Practices Observed

1. Clear commit messages with context
2. Comprehensive test coverage for critical paths
3. TypeScript strict mode compliance
4. ESLint configuration enforced
5. Proper error handling and validation
6. Component prop documentation



Areas for Improvement

1. Test coverage trending below 70% - **Action:** Add tests for utility functions
2. Missing API documentation - **Action:** Add JSDoc comments to endpoints
3. Some components need refactoring - **Action:** Break down large components
4. E2E test coverage sparse - **Action:** Implement Playwright tests

Review Checklist Template

- ☐ Code follows project style guide
- ☐ Tests added/updated for new features
- ☐ Type safety: TypeScript strict mode passes
- ☐ No security vulnerabilities detected
- ☐ Performance impact assessed
- ☐ Accessibility standards met (WCAG 2.1 AA)
- ☐ Documentation updated
- ☐ Dependencies reviewed (no unmaintained packages)
- ☐ Commit messages are clear and descriptive
- ☐ No console errors or warnings
- ☐ Backwards compatibility maintained

Next Steps

1. **Increase Test Coverage** to 75%+ target
2. **Document API Endpoints** with OpenAPI/Swagger
3. **Implement E2E Tests** using Playwright
4. **Security Audit** quarterly scheduled
5. **Performance Monitoring** baseline established

Generated by: CI/CD Review Process

Review Period: Q2 2026